

RobustGaSP: Marginal posterior mode estimate vs maximum likelihood estimation

1D test example

We have a 1-dimensional input, one output, and 5 noise-free observations, obtained using the function $f(x) = 2 \sin(3x - 0.5)$:

$$x = \{0.1, 0.3, 0.5, 0.7, 0.9\}$$
$$y = \{-0.4, 0.78, 1.68, 2, 1.62\}$$

which is plotted in Figure 1.

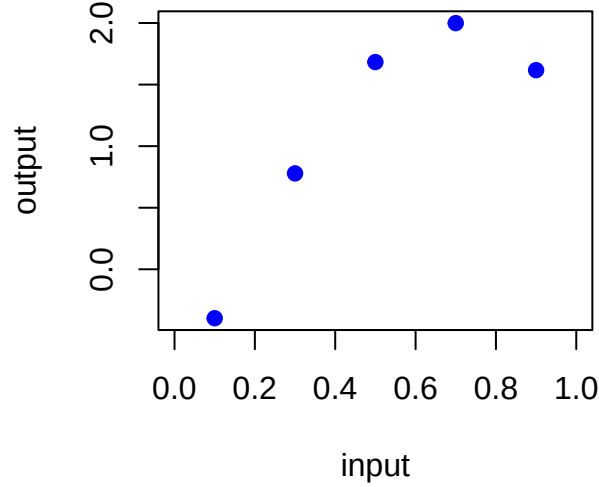


Figure 1: 1D example observations

Going forward, we will assume a linear mean function of the form

$$m(x) = \beta_0 + \beta_1 x.$$

In RobustGaSP's `rgasp()` function, this involves specifying the `trend` matrix H :

```
##      [,1] [,2]
## [1,]    1  0.1
## [2,]    1  0.3
## [3,]    1  0.5
## [4,]    1  0.7
## [5,]    1  0.9
```

as well as setting `zero.mean` equal to "No". The Gauss (squared exponential) *and* Matern 5/2 kernels will be compared for both estimation methods. The output from these four `rgasp` calls is shown below, first using maximum likelihood estimation for the Gauss and Matern 5/2 kernels respectively, and second using marginal posterior mode estimation for the Gauss and Matern 5/2 kernels respectively.

```

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 1
## The value of the profile likelihood function is 0.00239245
## Optimized range parameters are 0.001124683
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.064
## Start of the optimization 2 :
## The number of iterations is 36
## The value of the profile likelihood function is 1.936771
## Optimized range parameters are 0.9805695
## Optimized nugget parameter is 0
## Convergence: FALSE

```

```

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 1
## The value of the profile likelihood function is 0.00239245
## Optimized range parameters are 0.001124683
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.064
## Start of the optimization 2 :
## The number of iterations is 12
## The value of the profile likelihood function is 0.1638173
## Optimized range parameters are 0.2830598
## Optimized nugget parameter is 0
## Convergence: TRUE

```

```

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 44
## The value of the marginal posterior function is -0.2280412
## Optimized range parameters are 3.655377
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.064
## Start of the optimization 2 :
## The number of iterations is 25
## The value of the marginal posterior function is 4.466659
## Optimized range parameters are 1.114853
## Optimized nugget parameter is 0
## Convergence: TRUE

```

```

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 21
## The value of the marginal posterior function is -5.227341

```

```

## Optimized range parameters are 619828629467
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.064
## Start of the optimization 2 :
## The number of iterations is 16
## The value of the marginal posterior function is 3.12738
## Optimized range parameters are 4.677144
## Optimized nugget parameter is 0
## Convergence: TRUE

```

To visualise the posterior mean functions returned by these four functions, predictions are made at 1000 new points on the range $[0, 1]$. Once again, **rgasp** requires specification of the ‘trend’ matrix H , this time containing the testing inputs.

These 1000 points are plotted in Figure 2 (red line), along with 95% credible intervals (red dashed line) along with the actual function (black line) and 5 observations (blue points).

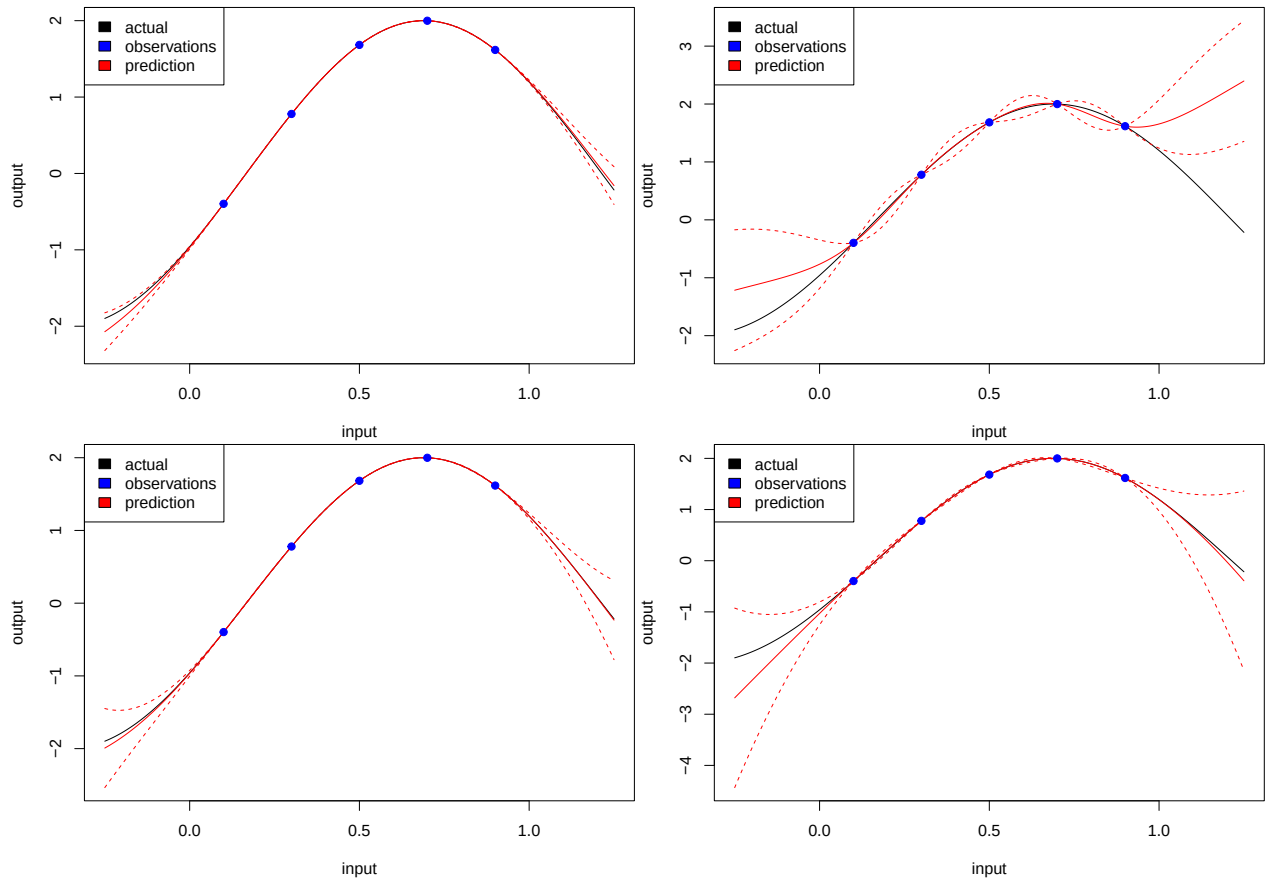


Figure 2: Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation for the Gauss (left) and Matern 5/2 (right) kernels.

Various checks are performed (as in the example from **RobustGaSP**’s Help page):

Table 1: RobustGaSP manual’s suite of checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
Maximum likelihood	Gauss	1	0.001	0.0739
Maximum likelihood	Matern 5/2	0.82	0.5	0.7133
Marginal post. mode	Gauss	1	0.0004	0.1577
Marginal post. mode	Matern 5/2	1	0.03	0.6329

Finally, in preparation for the real world data where we cannot perform out-of-sample validation, we’ll perform leave-one-out validation: for each of the observations, fitting a GP using all observations except this one, then using the fitted emulator to predict the output at the excluded observation’s input setting. The 95% credible interval for each prediction is checked against the actual output value, and the proportion of these CI’s that contain the respective observation is found. The results are shown in Tables 2-5 below.

Table 2: Maximum likelihood with Gauss kernel (60% validation)

actual	loo_prediction	lower95	upper95	validated
-0.40	0.81	0.18	1.44	0
0.78	0.54	-0.16	1.23	1
1.68	1.00	0.22	1.77	1
2.00	1.65	1.17	2.14	1
1.62	3.04	2.61	3.47	0

Table 3: Maximum likelihood with Matern 5/2 kernel (60% validation)

actual	loo_prediction	lower95	upper95	validated
-0.40	0.81	0.18	1.44	0
0.78	0.54	-0.16	1.23	1
1.68	1.00	0.22	1.77	1
2.00	1.65	1.17	2.14	1
1.62	3.04	2.61	3.47	0

Table 4: Marginal post. mode with Gauss kernel (80% validation)

actual	loo_prediction	lower95	upper95	validated
-0.40	-0.58	-0.65	-0.51	0
0.78	0.82	0.78	0.86	1
1.68	1.67	1.62	1.72	1
2.00	2.00	1.87	2.12	1
1.62	1.78	1.14	2.43	1

Table 5: Marginal post. mode with Matern 5/2 kernel (80% validation)

actual	loo_prediction	lower95	upper95	validated
-0.40	-0.58	-0.65	-0.51	0
0.78	0.82	0.78	0.86	1
1.68	1.67	1.62	1.72	1
2.00	2.00	1.87	2.12	1
1.62	1.78	1.14	2.43	1

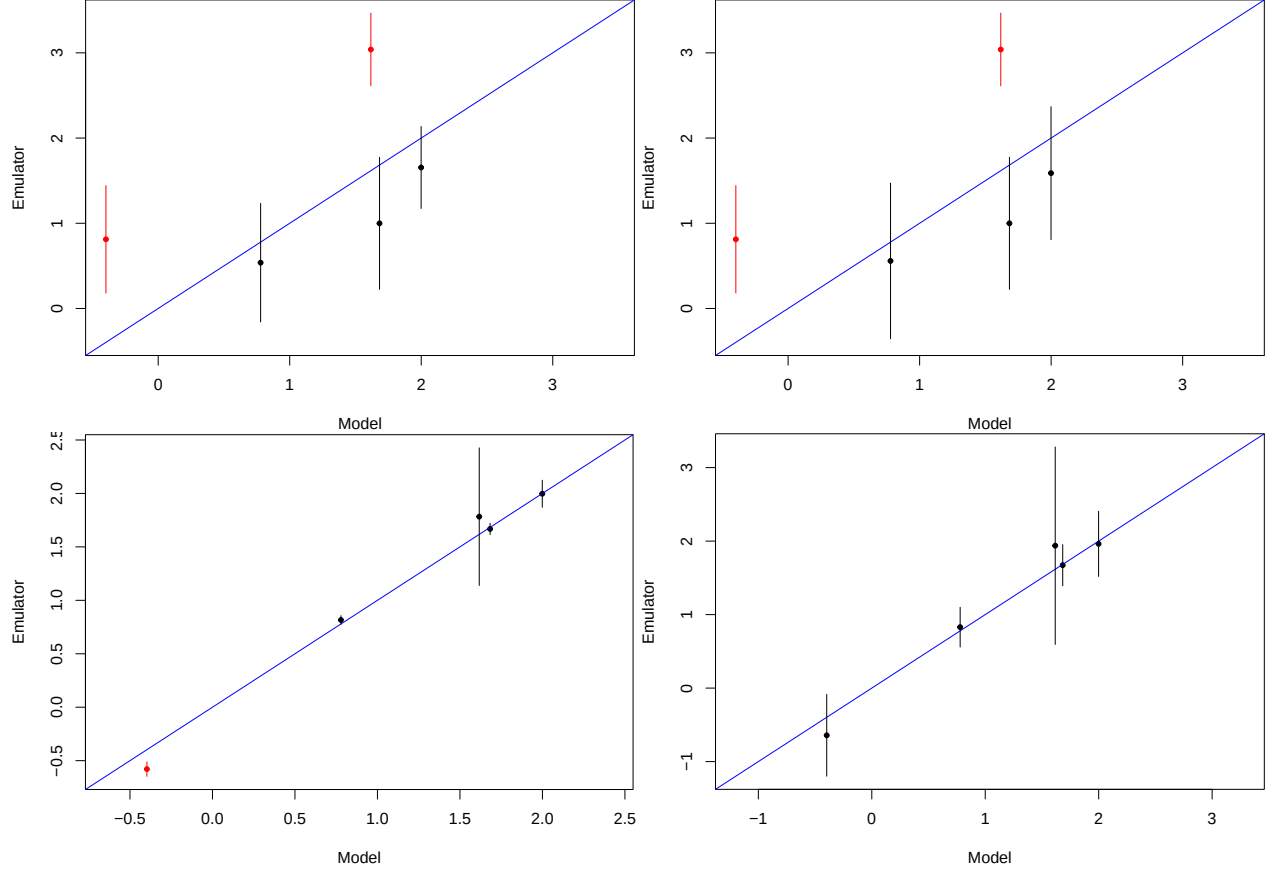


Figure 3: LOO cross validation. Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation for the Gauss (left) and Matern 5/2 (right) kernels.

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel. The estimates returned by both estimation methods using the Gauss kernel are shown in Table 6.

Table 6: Estimates table

Estimation	Kernel	Trend 1	Trend 2	Range	Variance
MLE	Gauss	-1.13	0.18	0.98	5.05
PM	Gauss	-0.7	-0.57	0.79	15.1

Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 1000 testing points by-hand, we require the matrix H containing the inputs for the 5 observations and well as matrix A , its inverse, and $\mathbf{t}(x^*)$ for all 1000 of the new inputs x^* . A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 4).

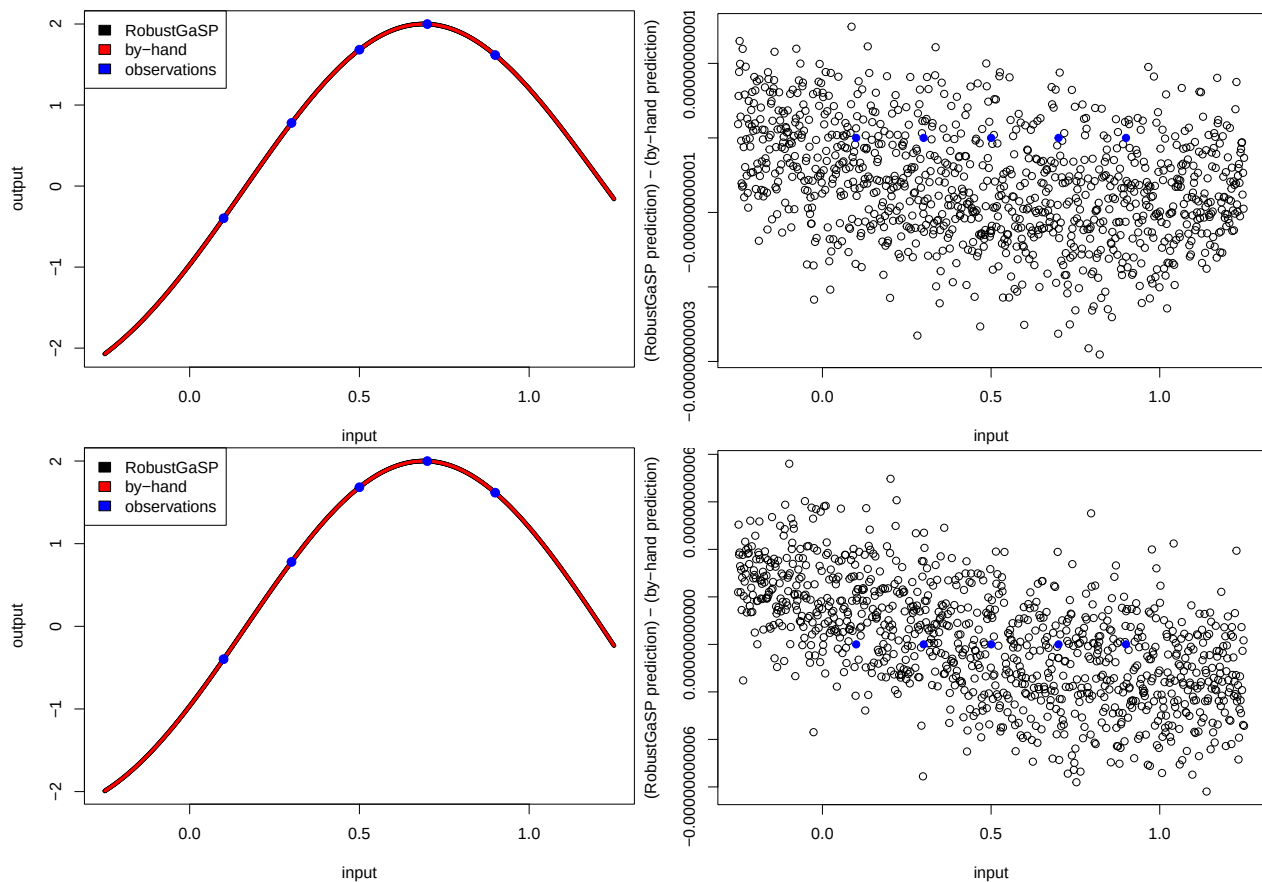


Figure 4: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE/ Bottom: RM.

For both we calculate

$$\frac{\sum_{i=1}^{1000} \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{1000}$$

then divide this by 10^{-8} (machine precision) to get 0.0008 for MLE and 0.002 for PM (the latter approximately a factor of 2 larger than the former).

Why don't RobustGaSP's estimates change? What is the stochastic element of DiceKriging?

Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

We check that the emulated derivative approximately matches the derivative of the actual function.

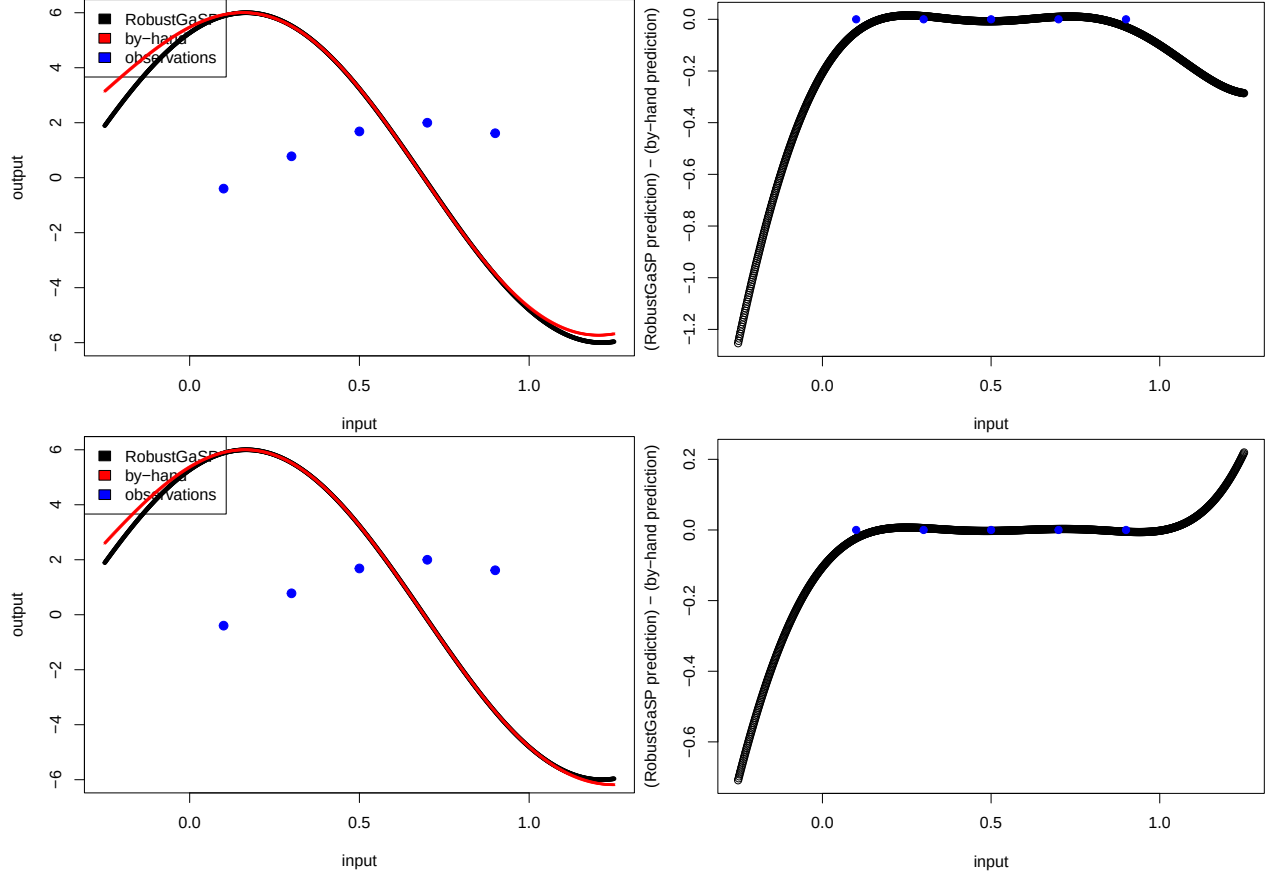


Figure 5: The estimation of the derivative found by each estimation method matches the true derivative well (left), with the latter minus the former found to be essentially zero (right). Top: DiceKriging. Bottom: RobustGaSP.